

Notation, conventions, and the running example

The symbols, the constants, and the 7B configuration that every chapter quietly assumes — collected on six pages so no chapter has to restate them

Every chapter of this book leans on a shared set of symbols, a shared bag of constants, and one model configuration that gets rebuilt, sharded, packed, and distilled from Chapter 3 to Chapter 30. Rather than restate them each time, they live here. If you open the book in the middle — which the preface encourages — this appendix is the two-minute orientation that makes any chapter readable.

B.1 Symbols

Table B.1 — Notation used throughout. Where the literature is split, the book picks one convention and holds it.

SYMBOL	MEANING	FIRST SERIOUS USE
N	model parameters (count)	Ch. 6
D	training tokens (count)	Ch. 6, 7
C	training compute, FLOPs — $C \approx 6ND$	Ch. 6
d	model (hidden) dimension, a.k.a. d_model	Ch. 3
d_{ff}	MLP inner dimension (SwiGLU: 3 matrices of it)	Ch. 3
L	layer count — <i>layers</i> , not <i>sequence</i>	Ch. 3
s	sequence length (context), in tokens	Ch. 8 on; Ch. 6 wrote it L once — this book's one regret
a	attention head count	Ch. 3, 8
b	microbatch size (sequences per device per step)	Ch. 8, 14
V	vocabulary size	Ch. 2, 11

SYMBOL	MEANING	FIRST SERIOUS USE
θ, W	parameters generically; a specific weight matrix	Ch. 1
$z, h, \hat{y}$	logits; activations; the model's prediction	Ch. 1
L or $\mathcal{L}$	the loss (context disambiguates from layer count)	Ch. 1, 2
η, α	learning rate (the RL chapters use α)	Ch. 4, 25
$\pi_\theta, \pi_{\text{ref}}$	policy; frozen reference policy	Ch. 24, 25
$\hat{A}$	advantage estimate (GRPO: group-normalized)	Ch. 25
G	rollout group size per prompt	Ch. 25
DP · TP · PP · EP	data / tensor / pipeline / expert parallel degrees	Ch. 14

B.2 The running example: "our 7B"

One configuration threads the whole book: a Llama-2-7B-shaped dense transformer, chosen because its numbers are round enough to compute by hand and its shape is close enough to production practice to matter. When a chapter says "our 7B," it means exactly this:

Table B.2 — The running configuration, and the numbers derived from it by hand.

QUANTITY	VALUE	DERIVED WHERE
layers L / hidden d / heads a	32 / 4096 / 32	Ch. 3
MLP d_{ff} (SwiGLU) / vocab V	11,008 / 32,000	Ch. 3
parameters N	6.74 B	counted by hand, Ch. 3
forward FLOPs per token @ $s=4096$	15.4 GF	Ch. 8 (§8.1)
training FLOPs per token (fwd+bwd)	$46 \text{ GF} \approx 6N + \text{attention}$	Ch. 8 (§8.2)
training states (16 B/param)	108 GB	Ch. 4, 8
activations, $b=1, s=4096$, FlashAttention	18 GB	Ch. 8 (§8.3)
healthy single-node throughput (8×H100)	$\sim 9.8 \text{ K tok/s/GPU} \approx 40\% \text{ MFU}$	Ch. 8 (§8.6)

The example language is Spanish wherever a language must be chosen — tokenizer walkthroughs, eval slices, the input sentences of the interactive animations — and the example datasets are FineWeb-Edu (sample-10BT) for pre-training text and UltraChat-200k for dialogue, as fixed in the preface.

B.3 The constants worth memorizing

Table B.3 — The book's working constants. Each is derived once and then used everywhere; the page cited is the derivation, not a bare assertion.

CONSTANT	VALUE	WHAT IT PRICES	DERIVED IN
forward cost	$2N$ FLOPs/token	2 FLOPs per weight per token	Ch. 6, 8
training cost	$6ND$ FLOPs	backward = 2× forward	Ch. 6, 8
attention surcharge	$\times (1 + s/6d)$	the pairwise term $6ND$ ignores	Ch. 6
recompute penalty	$6N \rightarrow 8N$	checkpoint-everything reruns the forward	Ch. 8
training state	16 bytes/param	bf16 w + g, fp32 Adam m, v + master	Ch. 4
activations/layer	$s b d (34 + 5as/d)$ bytes	the quadratic term FlashAttention deletes	Ch. 8
H100 ridge	≈ 295 FLOPs/byte	989 TF/s ÷ 3.35 TB/s: compute- vs memory-bound	Ch. 8
MFU	$6N \cdot \text{tok/s} / \text{peak}$	the honest utilization number (dense peak!)	Ch. 8
inference cost	$2N$ FLOPs/token	one forward, no backward	Ch. 6, 7
Chinchilla ratio	$D \approx 20N$	compute-optimal tokens — a starting point, bent by inference (Ch. 7)	Ch. 7

B.4 Conventions

Cross-references "§14.3" point within the book; "Ch. 14" names the chapter. Numeric claims carry footnotes to primary sources; our own measurements are marked as ours and rounded. Every figure is grayscale-plus-navy inline SVG with a numbered caption; code blocks show the lines that matter and point to the companion repository for the rest. Benchmarks are always reported with their harness config per Chapter 9 — a score without its template, shot count, and

normalization is treated as a number without units, in this book and in the reader's future reports, which is rather the point of the whole exercise.

Referenced implicitly from every chapter; explicitly from the preface. Corrections: the errata file in the course repository.